

# Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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6 September 2026

## Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Its demonstrated results establish what coordination leaves undetermined: increasing coupling, decreasing diffusion and a matched common drive can produce identical stationary phase distributions; equal phases permit incompatible local states; and a plasticity control preserves coherence while losing alignment with a specified environmental template. The conditional construction glues compatible spatial measures into one state and makes that state the unique fixed point of a contracting reconstruction map. Lean 4 checks the composition under eight connecting hypotheses and additional structures, with compatibility and accuracy at the glued state supplied as premises. We propose the glued state as a formal correlate of conscious unity and its accurate internal reconstruction as a correlate of a minimal self. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response and the distributional

equivalence requires measurements that distinguish recovery mechanisms. Testing the account further requires identifying neural content and self-encoding independently, then determining whether coordination maintains their agreement and whether these relations matter for unified experience.

# 1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information about where the cup is while contributing different contents to the episode.

19 Synchronous activity in the two processes would leave any disagreement  
20 about that shared quantity unresolved. The formal counterexample isolates  
21 this problem using spatial mass profiles at a common phase that disagree  
22 on an overlap. We return to the cup example to ask what agreement would  
23 mean for perceptual content, and what an internal representation would have  
24 to add to place the perceiver in that situation. It is an illustration of the  
25 modelling task, not a neural model implemented here.

26     Awakening gives the physical requirement a concrete setting. We hypoth-  
27 esize that changing extracellular geometry slowly increases effective coupling  
28 through a synchronization threshold. Under stationary branch tracking, a  
29 noisy phase model fixes a square-root onset of coherence. The experiment  
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a  
31 fitted onset exponent depends on the drive. Moreover, successive stationary  
32 phase distributions can be identical under increasing coupling, decreasing  
33 diffusion or a suitably matched common drive. Recovery must therefore be  
34 measured in a way that identifies dynamics and distinguishes causes, rather  
35 than inferred from the shape of a coherence trace alone.

36     A plasticity control tests another route to representation. Its squared-  
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster  
38 model favors connections across the clusters that define the diagnostic tem-  
39 plate. Coherence survives while alignment with that template declines. To-  
40 gether with the overlap counterexample, this result establishes a limit on  
41 what the coordination model explains. Compatible local content needs its

own mechanism and observable.

The contribution has three levels. The separation results establish limits on what coordination explains. The conditional construction gives a unique global state from specified compatible spatial measures and makes it the unique fixed point of a contracting map assumed to reconstruct that state. Lean 4 checks the composition, and toy witnesses establish joint satisfiability of its assumptions. The biological interpretation proposes that the glued state and its internal reconstruction capture conscious unity and a minimal self. This last step requires identifying meaningful neural descriptions and explaining how coordination maintains their agreement. Figure 1 locates the proposed connections, and Section 6 states the physical and thermodynamic assumptions.

## 2 Coherence: the coupling model

### 2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[ \omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here  $D > 0$  is phase diffusion,  $W_i$  are independent Wiener processes, and  $K_{ij}$  has inverse-time units, with angles dimensionless. The uniform mean-field convention is  $K_{ij} = K/N$ . Frequency heterogeneity and phase diffusion are

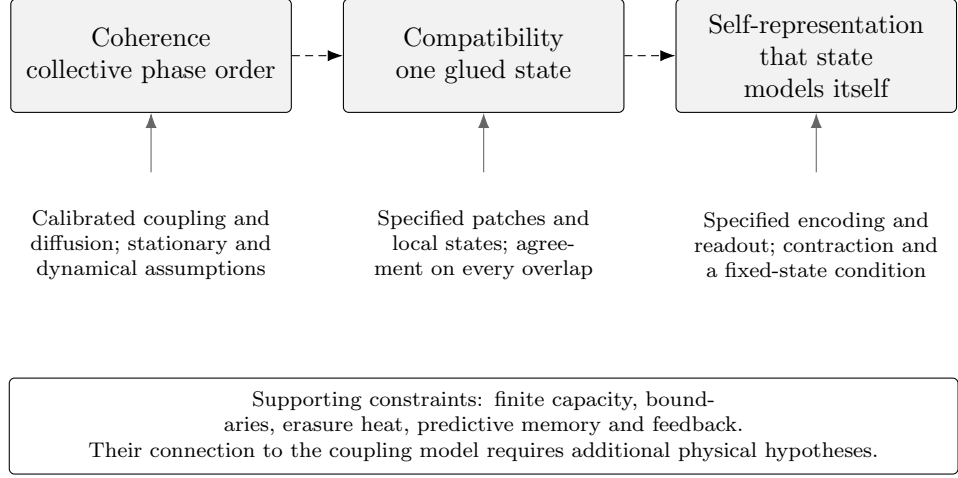


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 separate parameters.

61 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

62 where  $I_n$  are modified Bessel functions and  $\psi$  is the collective phase. Lean  
 63 verifies normalization, the circular order parameter and self-consistency for  
 64 this density. At  $K \leq K_c = 2D$ , the only nonnegative self-consistent order  
 65 parameter is zero. At  $K > K_c$ , exactly one positive solution coexists with

66 zero; it emerges continuously and increases strictly with  $K$ . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 These results characterize the stationary branch (Fig. 2). Connecting it  
 68 to finite trajectories requires a dynamical mean-field limit and dynamical  
 69 selection, neither proved here. Propagation of chaos concerns the first of  
 70 these steps (Sznitman, 1991); exchange symmetry alone does not establish  
 71 independence, especially when realizations have an unpinned collective phase.

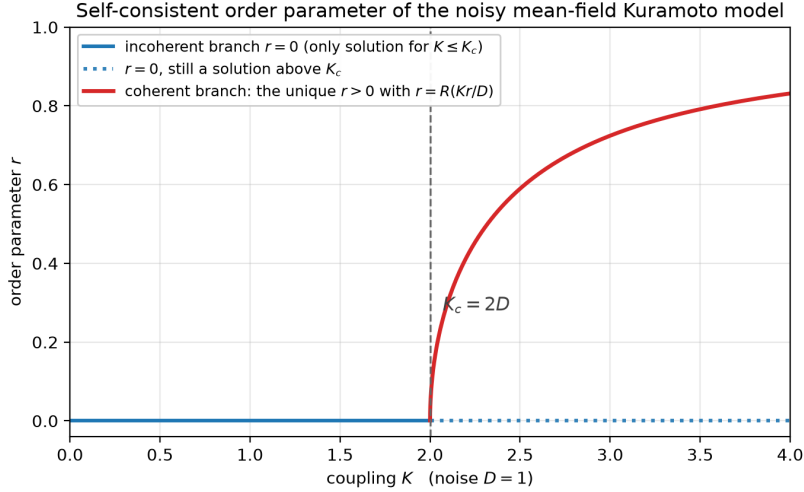


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion  $D = 1$ . The positive coherent branch appears continuously above  $K_c = 2D$  and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

72 A spatial version replaces the finite sum by an integral against a kernel  
 73  $K(x, y)$ . The substrate measure has total mass one, making the mean-field

74 strength a kernel average. Applying the scalar threshold to a heterogeneous  
 75 or signed kernel requires a justified reduction to that mean-field description.

## 76 **2.2 Physical identification and calibration**

77 The cortical hypothesis identifies part of the effective coupling with endoge-  
 78 nous extracellular electromagnetic interactions. Fields can modulate spike  
 79 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field  
 80 models have been proposed for coordinating distributed neural information  
 81 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is  
 82 whether that interaction supplies sufficient coupling relative to diffusion.  
 83 Changes in phase organization under propofol (Bastos et al., 2021) moti-  
 84 vate studying this regime, but do not measure the coupling itself.

85 The physical field predicate requires a jointly continuous kernel, a nor-  
 86 malized substrate, a mean-field strength equal to the spatial average, and no  
 87 single site providing more than half that strength. Regular mesh refinement  
 88 establishes convergence of a scalar coupling-energy quadrature; constructing  
 89 the kernel and relating it to trajectories are separate modelling tasks. The  
 90 formal witnesses supply toy instances of the predicate.

91 Calibration is essential because a spike-time shift is not a coupling rate.  
 92 An estimate proportional to  $N\Delta tf$ , using population size, timing shift and  
 93 oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

94 requires a rate  $\gamma$ , where  $s$  is the timing shift per unit field and  $E$  the field  
 95 amplitude. This rate specifies how induced timing shifts persist and accumu-  
 96 late into phase drift. Supplemental Material explains why the other factors  
 97 cannot determine it. The biological threshold test requires independent es-  
 98 timates of the interaction kernel, its aggregation, and phase diffusion in the  
 99 same cortical regime.

### 100 **3 Testing coherence through recovery after** 101 **awakening**

102 The proposed control variable is extracellular geometry. Sleep-associated  
 103 changes in extracellular space and clearance (Xie et al., 2013) motivate a  
 104 coupling parameter that evolves more slowly than phase dynamics. The  
 105 extracellular volume fraction is approximately  $\alpha = 0.2$  at diffusion tortuosity  
 106  $\lambda \approx 1.6$  (Syková & Nicholson, 2008); the cited sleep study reports a rise  
 107 in  $\alpha$  from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical  
 108 tissue conducts approximately resistively at  $0.3\text{--}0.6\text{ S m}^{-1}$  (Barbour, 2017).  
 109 In an effective-medium description at fixed transmembrane current density,  
 110 a larger extracellular volume fraction raises conductivity and reduces field  
 111 amplitude.

112 This gives the coupling-gated hypothesis a directional prediction: contrac-  
 113 tion of extracellular space on awakening raises  $E$  and, with the other factors  
 114 in Eq. (4) held fixed, raises effective coupling. The proposed geometry–

115 coupling relation and its role in human recovery both require measurement.  
 116 The effective-medium argument supplies a sign under stated conditions; cal-  
 117 ibration supplies the magnitude and the mapping to the scalar model.

118 If  $K(t)$  rises through  $K_c$  and the phase distribution tracks the coherent  
 119 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

120 The crossing time satisfies  $K(t_0) = K_c$ . It is a prediction of latency only when  
 121 coupling is measured independently of coherence. The subsequent shape  
 122 and saturation depend on the later coupling trajectory. Critical slowing  
 123 down restricts branch tracking near threshold, while finite populations have  
 124 fluctuating order below it. A measurable onset therefore needs an explicit  
 125 criterion above that fluctuation floor.

### 126 **3.1 What a recovery distribution can identify**

127 Equation (2) gives a concentration–coherence relation with no fitted shape  
 128 parameter. Testing it requires separate estimators:  $r$  from circular resultant  
 129 length and  $a$  from density shape. Estimating  $a$  by inverting  $R$  from the  
 130 measured  $r$  makes agreement automatic. Even a valid agreement tests a  
 131 statistical description shared by the von Mises family; it does not determine  
 132 the coupling medium.

133 There is a stronger limitation. Successive equilibrated phase distributions

134 cannot, by themselves, distinguish three different recovery mechanisms. Let  
 135  $q(t)$  be a prescribed ratio crossing 2, let  $r(q)$  denote its stationary branch,  
 136 and choose a fixed rate  $D_0 > 0$ . Increasing coupling with  $K(t) = D_0 q(t)$   
 137 and  $D = D_0$  gives the same stationary density as fixed coupling  $K_0$  with de-  
 138 creasing diffusion  $D(t) = K_0/q(t)$ . Both have concentration  $a = q(t)r(q(t))$ .  
 139 Uncoupled oscillators under a common prescribed drift  $-h(t) \sin(\theta - \psi)$ , with  
 140  $h(t) = D_0 q(t)r(q(t))$  and  $D = D_0$ , have that same density. Their distribu-  
 141 tions coincide window by window, even though their physical causes differ.  
 142 Supplemental Material supplies the construction and its numerical feasibility  
 143 check.

144 The missing observable is the physical clock. Resolved phase increments  
 145 contain rate information: in the ideal continuous-path model, quadratic vari-  
 146 ation accumulates at rate  $2D$ , while conditional increment means constrain  
 147 drift. This can separate coupling and diffusion under the specified observa-  
 148 tion model. Distinguishing endogenous coupling from the matched common  
 149 drive additionally requires a measured drive or a controlled perturbation, be-  
 150 cause their mean-field drifts can coincide. Identifying the coupling as electro-  
 151 magnetic then requires evidence specific to that interaction. This hierarchy  
 152 determines the experiment: distribution, dynamics and physical mechanism  
 153 are distinct targets.

## 154 **3.2 Finite ramps change the observable onset**

155 Finite simulations characterize how the stationary prediction can appear in  
156 a recovery measurement. The ramp study uses  $N = 2000$ ,  $D = 1$  and 32  
157 replicas at four coupling speeds. Escape is the first of three consecutive  
158 saved samples at  $r \geq 0.2$  after threshold. The supplement gives parameters,  
159 estimators and further controls.

160 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the  
161 fastest window. The three slower ramps have mean coupling delays 0.1947,  
162 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,  
163 compared with 0.443 to 0.448 from the same estimator on the exact stationary  
164 branch at the same coupling grids. At the fastest speed the ensemble mean  
165 never reaches the declared fit window; the returned value 0.964 is a local  
166 log-slope of a delayed trajectory.

167 This is delayed passage through a noisy bifurcation (Berglund & Gentz,  
168 2002), with a direct consequence for the awakening test: the apparent expo-  
169 nent depends on the ramp and on the fitting window. A deviation from the  
170 square-root law constrains the adiabatic approximation only where branch  
171 tracking is independently justified. The sweep does not bracket the specified  
172 deviation-based critical ramp speed.

173 Additional controls test the approximations used by the physical model.  
174 Time-step refinement attributes most of the sampled discrepancy from the  
175 stationary order to integration bias. Population-size diagnostics show why  
176 a shared random collective orientation can preserve unconditional pair cor-

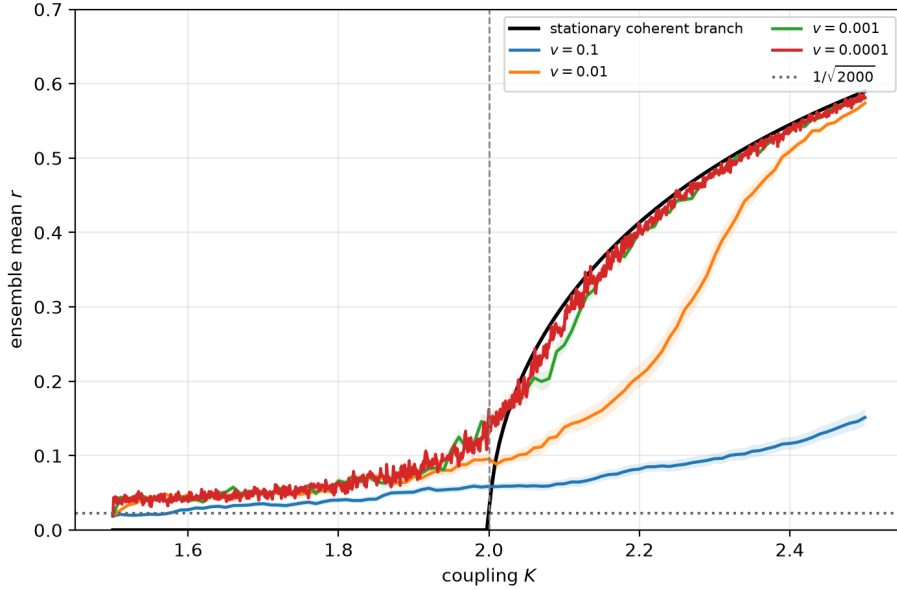


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds  $v$ , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

177 relation in the coherent regime. Spatial and signed-network controls exhibit  
 178 coherence at specified total coupling, with the refined signed-network cross-  
 179 ing bracket  $K_{\text{eff}}/D \in [1.958, 2.009]$  still containing 2. These controls delimit  
 180 finite regimes and numerical resolution; the supplement reports their full  
 181 evidence. Calibrating the cortical coupling remains the experimental task.

### 182 3.3 What the estimators produce on EEG

183 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory  
 184 exercise of the concentration and coherence estimators. The implemented

185 analysis pools channels over 100-ms windows, estimating a spatiotemporal  
 186 phase distribution. Bipolar and common-reference montages produce differ-  
 187 ent residuals. The bipolar results are compatible with the Bessel relation,  
 188 but the concentration range is too narrow to distinguish it from its linear  
 189 approximation at the observed scatter.

190 An informative recording needs a broader concentration range, estima-  
 191 tor calibration that accounts for dependence, and an observation model for  
 192 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-  
 193 esis remains untested by these sedation recordings.

### 194 **3.4 A protocol for the awakening experiment**

195 The analysis defines six requirements for the physical test.

- 196 **1. Coupling and diffusion in the same units.** Estimate  $K$  and  $D$   
 197 from resolved phase dynamics in the same cortical regime, separating  
 198 diffusion, frequency heterogeneity and observation noise. A measured  
 199 drive or controlled perturbation is additionally needed to distinguish  
 200 endogenous coupling from the matched common-drive alternative.
- 201 **2. Geometry calibrated to coupling.** Establish the relation between  
 202 an extracellular-geometry observable and effective coupling, with un-  
 203 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-  
 204 quires its own validation; the conductivity argument supplies only the  
 205 expected sign.

206     **3. Adequate phase measurements.** For the proposed spatial estima-  
207     tor, use at least 100 sites per estimate,  $r$  from resultant length, and  
208      $a$  from unweighted regression of log density on the cosine of centered  
209     phase. This numerical minimum needs null calibration at the actual  
210     sample count, concentration range and dependence. Temporal pooling  
211     changes the estimand.

212     **4. An independent recovery endpoint.** Measure behavioral recovery  
213     separately from the phase trace.

214     **5. A predeclared onset criterion.** Fix a coherence threshold above the  
215     finite-size floor and account for finite ramp speed before interpreting  
216     an onset exponent.

217     **6. Matched model comparison.** Compare coupling-gated recovery  
218     with diffusion change, common drive and alternative slow-recovery  
219     models under the same observation and filtering assumptions. Station-  
220     ary distributions alone cannot resolve the equivalence in Section 3.1.

221     Failure of the concentration–coherence relation over an informative range  
222     would constrain the phase-density model. Failure of the calibrated geometry–  
223     coupling association would constrain the proposed physical gate. A com-  
224     peting dynamical explanation that better predicts independently measured  
225     recovery would count against its role in awakening. These tests address the  
226     first requirement of the framework. Even a favorable outcome leaves the  
227     question of what the coordinated activity represents.

## 228 4 Compatibility: from coordinated activity 229 to a global description

### 230 4.1 Why common timing is insufficient

231 Consider overlapping patches  $U$  and  $V$ , each carrying a local description.  
232 Compatibility requires that restricting either description to  $U \cap V$  gives the  
233 same object. Agreement concerns the shared domain; descriptions on distinct  
234 parts can remain different. For the finite spatial measures used here, equal  
235 overlap restrictions give a unique measure on the covered domain. This is  
236 the gluing property formalized by a sheaf.

237 In the cup example, suppose a visual description locates the cup on a  
238 table and a reaching description relates the hand to the cup. To compare  
239 them, a model must express their shared quantity, such as cup location, in  
240 a common coordinate system. Restriction would retain that quantity while  
241 dropping details peculiar to each description. Agreement on it permits the  
242 visual description to contribute surface appearance and the reaching descrip-  
243 tion to contribute hand configuration without duplicating the whole situa-  
244 tion. Disagreement after coordinate conversion would make the two estimates  
245 incompatible at that level of description, however closely timed their activity.

246 The formal counterexample makes the additional requirement visible. Ev-  
247 ery patch has the same oscillator phase, yet two local mass profiles assign  
248 different densities to a shared site. They cannot be restrictions of a single

249 global measure. Phase order therefore cannot establish compatibility. Con-  
250 tinuity of a physical field does not repair disagreement in the descriptions  
251 associated with it.

252 For a theory of experience, the attraction of gluing is this contribution  
253 of different local contents to one description. The cup example does not yet  
254 instantiate the theorem: receptive territories and content domains must be  
255 related to the spatial patches, and their descriptions must have the required  
256 extension property. In particular, a decoder assigning probability laws to per-  
257 ceptual contents needs a further extension model, because matching overlap  
258 marginals need not determine a unique global joint distribution (Abramsky &  
259 Brandenburger, 2011). The spatial-measure construction provides a precise  
260 case of gluing; a content model must justify its own existence and uniqueness  
261 claims.

## 262 **4.2 Coherent adaptation can lose environmental align-** 263 **ment**

264 An adaptive coupling rule might appear to supply that missing representa-  
265 tional structure. The squared-drift control shows why the objective mat-  
266 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel  
267 when summed over nodes. Minimizing the sum of squared drifts at that  
268 fixed mean therefore rewards equalizing drift across nodes. In the chosen  
269 frequency-cluster model, this favors coupling between unlike-frequency clus-

ters, whereas the diagnostic template rewards coupling within like-frequency clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[ \omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator  $A$  and its adjoint, obtains  $\nabla\sigma = (2/D)A^*(AK + \omega)$ , and proves descent for the specified gradient flow. This is a squared-drift functional; identifying thermodynamic entropy production would require probability-current information.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. The frequency-cluster template is a diagnostic covariance of shared latent signals, not a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-gradient and frozen-kernel controls, including comparisons at overlapping deformation and coherence ranges and against partitions blind to frequency.

The objective reduction is transient: averaging over complete update intervals removes the apparent gain seen immediately after updates. A pre-specified grid of 6 rate–cadence combinations contains no case meeting the declared 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified

290 structure, with no demonstrated sustained cost reduction.

291     Kernel alignment and overlap compatibility are different observables.  
292 This control tests the narrower suggestion that coherent adaptation under  
293 this objective acquires the designated environmental structure. It fails for  
294 an intelligible reason: the objective rewards a different organization. A  
295 thermodynamically specified learning model can include data in its objective  
296 (Corberi et al., 2026); explaining agreement about the cup would likewise  
297 require a learning rule connected to the relevant content.

### 298 **4.3 What must be specified and measured**

299 Compatibility enters the construction as a hypothesis about the local states.  
300 For deterministic positive coupling, convergence to a potential minimum is  
301 proved under an explicit initial basin condition; the proposed bridge E78  
302 connects supercritical noisy coherence to relaxation of the specified cover.  
303 Neither step explains how initially inconsistent representations become con-  
304 sistent. That is the mechanism a biological account must add.

305     A neural test must first identify the domains and objects whose agreement  
306 is at issue. Two decoders should estimate a shared content on overlapping  
307 domains, with restriction maps specified before inspecting their agreement.  
308 Their discrepancy needs a null under decoding noise, and the procedure must  
309 distinguish deliberately constructed compatible and incompatible examples.  
310 A common prior can make separate decoders agree artificially; region-specific  
311 bias can make them disagree artificially. The supplement develops these re-

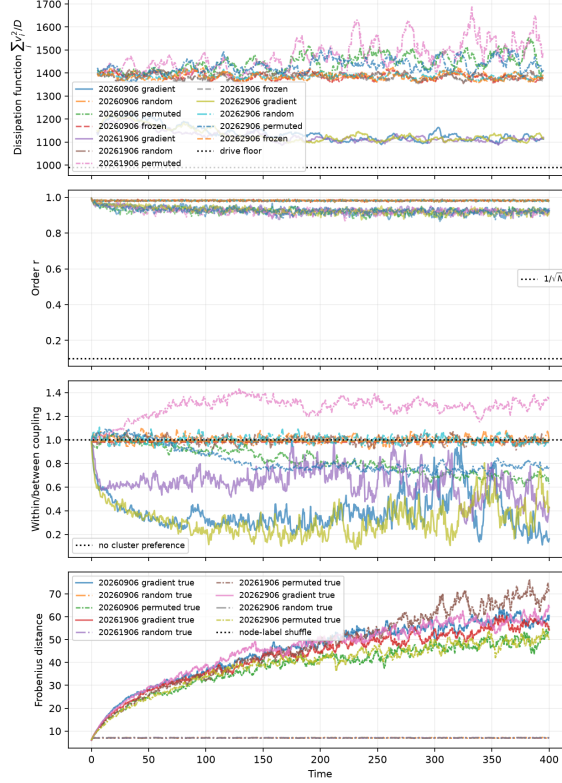


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

312 quirements. An estimator built from phase agreement would miss the prop-  
 313 erty the counterexample separates.

314 Choosing a cover is also part of choosing a candidate subject. Trivial  
 315 descriptions can agree while carrying little content, and a suitably chosen  
 316 partition can conceal disagreement. The scientifically useful choice must  
 317 therefore be constrained independently by what local activity encodes and  
 318 which shared quantities can be compared. The uniqueness theorem then  
 319 answers a precise question about that choice: whether the supplied local  
 320 descriptions determine one global state. It leaves the empirical selection of  
 321 domains and descriptions to the model.

## 322 **5 Self-representation: a state that models it-** 323 **self**

### 324 **5.1 The reflexive construction**

325 Compatibility yields a global state. Reflexivity adds a relation between that  
 326 state and an encoding within it. Equip global sections with a complete  
 327 metric. Under the additional restriction-resonance condition, the encoding  
 328 of a section  $s$  is its restriction to a region  $U$ , and the map takes the form

$$F(s) = \text{readout}(s|_U). \tag{7}$$

329 The same region and readout are used across the model’s state space. If  $F$   
330 contracts and fixes the section  $s_*$  supplied by the cover, Banach’s theorem  
331 makes  $s_*$  its unique fixed point (Banach, 1922). The formal bridge E89 sup-  
332 plies the contraction law and  $F(s_*) = s_*$  for the boundary’s encoding–readout  
333 composite. Identifying that encoding with restriction is a separate resonance  
334 condition; E89 does not supply it. The supercritical order parameter pro-  
335 vides an inequality used by the specified contraction bound; constructing the  
336 encoding and readout remains a separate obligation.

337 The equality ties the reconstruction to the state formed from the local  
338 descriptions. In the cup example, the candidate global state would include  
339 the represented scene and the perceiver’s relation to it. A local encoding  
340 would have to retain enough information for the specified readout to recover  
341 that modelled state. An encoding of cup position alone would be insufficient  
342 if the readout could not recover the represented hand and viewpoint. The  
343 term *predictive* names this reconstruction map; a temporal neural prediction  
344 requires a dynamical interpretation.

## 345 **5.2 Why connect these properties to unity and a min-** 346 **imal self?**

347 We identify the section glued from compatible local states with the unity  
348 of a conscious episode, and its accurate internal reconstruction with a mini-  
349 mal self. Two features of the episode motivate this interpretation: different

350 contents belong to one situation, and that situation is experienced from a  
351 perspective.

352     The unity proposal starts from a distinction between having several con-  
353 tents and having several experiences. Seeing the cup’s surface and feeling  
354 the reaching hand contribute different contents to one situation. Compat-  
355 ibility expresses their agreement about what they share while preserving  
356 those differences. An episode can also contain uncertainty or conflicting  
357 judgments. A model might represent the cup’s apparent position separately  
358 from a judged position and consistently describe the discrepancy. Deciding  
359 whether two estimates concern the same quantity or distinct aspects of ex-  
360 perience is therefore part of defining the content model. A choice made only  
361 to ensure agreement would give the unity proposal no empirical force.

362     The motivation for a minimal self is that the cup is encountered from a  
363 location and in relation to a body that can reach it. The proposed internal  
364 representation concerns the perceiver’s condition in that encounter: where  
365 the perceiver is situated, what is being represented and how the body relates  
366 to it. Such a representation could be implicit in the organization of activity;  
367 it need not describe a person verbally or carry an autobiographical narrative.  
368 Requiring it to reconstruct the same global state links the perspective to the  
369 episode whose contents it organizes.

370     The fixed-point condition requires accuracy about the very state contain-  
371 ing the encoding. Reconstruction alone is insufficient for a perspective: a  
372 recording can be reconstructed without being used by a system to represent

373 its own condition. The proposed interpretation therefore needs evidence that  
 374 the encoding helps the system distinguish changes in itself from changes in  
 375 its surroundings. Changing the encoded hand or viewpoint while holding  
 376 the external scene fixed should alter the system’s estimates of its relation  
 377 to the cup as the readout predicts. This functional criterion goes beyond  
 378  $F(s_*) = s_*$ ; whether the represented perspective is experienced remains a  
 379 further question.

380 Accuracy concerns the variables represented by  $s_*$ , rather than every mi-  
 381 croscopic physical fact. A cortical realization must specify those variables and  
 382 justify the required fidelity. Approximate self-models, errors and changing  
 383 states may require a weaker condition than the exact fixed point established  
 384 here.

385 We propose reflexivity as necessary because even an unreflective episode  
 386 has a perspective: the cup appears somewhere relative to the perceiver,  
 387 whether or not the perceiver attends to themselves. Requiring internal self-  
 388 encoding on that basis is the philosophical hypothesis. Gluing alone does  
 389 not require it; the dissociations below test the two commitments separately.

### 390 **5.3 Which cover and which self-model?**

391 Both uniqueness results are conditional on their objects. Gluing gives one  
 392 global extension of specified compatible local data; Banach gives one fixed  
 393 point of a specified contracting map. These solve ambiguity within a chosen  
 394 construction. They do not choose between different covers, different encod-

395 ings or different candidate subjects. Integrated information theory makes  
396 exclusion an explicit part of its account (Tononi et al., 2016); the present  
397 uniqueness results do not settle that selection problem.

398 The relevant cover and map must be tied to neural organization by evi-  
399 dence independent of the desired conclusion. Local descriptions should pre-  
400 dict the contents attributed to their domains, and their restrictions should  
401 preserve meaning on shared domains. The encoding and readout should be  
402 specified before testing them, with their sensitivity to changes in the pro-  
403 posed state measured over a declared range.

404 A constant readout returning one predetermined state contracts and has a  
405 fixed point, even when it discards an encoding supplied by restriction. Such a  
406 construction fails the sensitivity requirement above. This is why selecting the  
407 map requires evidence about its functional role in addition to mathematical  
408 existence.

## 409 **5.4 Dissociations that would bear on the identification**

410 A test of the full commitment must measure its clauses separately: effective  
411 coupling relative to diffusion, overlap compatibility, and the accuracy of the  
412 proposed self-encoding, alongside an independent endpoint for experience.  
413 A case meeting all three physical and representational conditions with evi-  
414 dence against a unified conscious episode would challenge their sufficiency.  
415 Unified experience with demonstrably incompatible local descriptions would  
416 challenge the unity identification. Unified experience with compatibility but

417 without the specified self-encoding would challenge the reflexive requirement.

418     Anesthetic emergence, dreaming and dissociative states are candidate set-  
419 tings because the framework asks whether coordination, unified content and  
420 self-representation can separate. Interpreting such evidence requires a de-  
421 clared relation between the experience endpoint and report; absence of report  
422 alone cannot define absence of an episode. The compatibility observable in  
423 the supplement is a specification, and the corresponding neural self-encoding  
424 has yet to be identified. These are the main obstacles to testing the inter-  
425 pretive commitment.

426     The proposal concerns which state is conscious. It supplies no explanation  
427 of why satisfying these relations should be accompanied by experience. Nor  
428 do its gluing and fixed-point conditions select an electromagnetic substrate:  
429 the cortical field commitment resides in the physical identification and cali-  
430 bration. Their multiple realizability leaves the status of other biological and  
431 artificial implementations open.

## 432   **6   Thermodynamic constraints and the con-** 433       **ditional composition**

434     The broader composition constrains one proposed physical implementation  
435 through finite-register capacity  $H(p) \leq \log |X|$ , erasure heat under a specified  
436 system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000), and a  
437 separate field-boundary result for disconnected vacuum components (Kibble,

1976). The supplement develops these results and their witnesses. The phase threshold and representational argument stand independently of a derivation from finite capacity.

The constraint most directly relevant to representation concerns predictive memory. For a state  $X_t$  driven by an environmental signal  $S_t$ ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (8)$$

Under the no-back-action Markov structure  $X_t \rightarrow S_t \rightarrow S_{t+1}$ , the bracket is nonnegative (Still et al., 2012). The bound limits nonpredictive memory at a given dissipation budget; zero retained information satisfies it too. Relating it to Landauer heat requires that memory waste be bounded by erased entropy at the same thermal scale and heat budget. These distinct information constraints (Rosas et al., 2026) leave content and learning dynamics to be specified.

An acting agent changes its future inputs, so the passive assumption requires justification for the variables and interval considered. A feedback model specifies a policy  $A_t \sim \pi(\cdot | X_t)$ , world dynamics  $S_{t+1} \sim K(\cdot | S_t, A_t)$ , an observation channel and a memory update. For finite states, the exact information balance is

$$I(X_t; S_t) - I(X_t; S_{t+1}) = I(X_t; S_t | S_{t+1}) - I(X_t; S_{t+1} | S_t). \quad (9)$$

Both conditional terms are nonnegative. The last term vanishes for the passive model; with feedback, the signed difference can be negative because actions create correlations with the future. Its causal interpretation requires an environmental state sufficient to account for the relevant dynamics.

The finite thermodynamic extension instead bounds joint entropy reduction. For an elementary update that holds the controller fixed, strictly positive forward and reverse path probabilities give  $\Sigma = D_{KL}(P_F||P_R) = \Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$  under local detailed balance, with  $Q$  positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa, 2013). This supplies the thermodynamic constraint for the active branch below. A proved two-substep extension composes actuation with a memory update initialized from its coordinate-swapped final law. It identifies the output with the perception–action channels and cancels the same intermediate entropy and expected energy. At a common thermal scale, the whole update’s joint entropy decrease is bounded by its total heat budget divided by  $k_B T$ , without assuming stationarity. The supplement gives the balances and reciprocal two-bit witnesses with explicit interaction energy, heat and work. Action and observation alias the existing states; additional physical registers require further accounting. A biological objective, learning rule and cortical action readout remain modelling choices.

## 475 6.1 What the composition establishes

476 Table 1 lists the connecting hypotheses of two branches. The theorem **chain**  
 477 uses passive predictive memory; **chain\_active** uses a specified finite feedback  
 478 process. Each accepts eight implications and shares the passage from coarse-  
 479 graining to the phase model and its proposed representational role. In both,  
 480 the finite register and the field boundary are independent starting objects.

481 For the active branch, the controller has the register’s state type and the  
 482 fourth node records

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (10)$$

483 where  $p, p'$  and  $\langle Q \rangle$  belong to that same controlled substep. E34Active sup-  
 484 plies positive path probabilities, local detailed balance at the register’s ther-  
 485 mal scale  $\theta$ , and the allocation  $\langle Q \rangle \leq B$ , with  $B$  equal to the register’s  
 486 heat. Landauer’s lower bound does not imply this upper-budget allocation  
 487 or identify erasure with actuation. E45Active requires the bound for these  
 488 specified data to place the coupling energies in the named convergent regime.  
 489 A coupling dynamics establishing that implication is an additional modelling  
 490 requirement.

491 Two structures also enter both theorems: a thermodynamic cover, in-  
 492 cluding overlap compatibility, and a reflexive boundary specifying the local  
 493 encoding and readout. Given these and the connecting hypotheses, both  
 494 conclude **UnifiedSelf**: the section glued on that cover is the unique fixed

| Connection      | Status                       | Required assumption                                                                                                                                                                                                                                                                |
|-----------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E12             | Independent physical premise | A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.                                                                                                                                                        |
| E23             | Bridge assumption            | The boundary is associated with a finite register whose update leaves states unreachable.                                                                                                                                                                                          |
| E34 / E34Active | Bridge assumption            | Passive: a predictive structure with the register’s thermal scale and heat budget, and nonzero wasted memory. Active: positivity, local detailed balance and the register’s heat allocation for the named elementary process. The proved two-substep bound supplies no allocation. |
| E45 / E45Active | Modelling assumption         | The branch’s bound, on predictive memory or the named feedback process, places coupling energies in the specified convergent coarse-graining regime.                                                                                                                               |
| E56             | Physical commitment          | A specified physical field realizes the limiting coupling and satisfies the kernel conditions.                                                                                                                                                                                     |
| E67             | Physical commitment          | That field operates above the synchronization threshold.                                                                                                                                                                                                                           |
| E78             | Bridge assumption            | Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.                                                                                                                                                                      |
| E89             | Modelling assumption         | A contracting map mediated by local self-encoding fixes the section glued on that same cover.                                                                                                                                                                                      |

Table 1: Connecting hypotheses of the passive and active compositions, eight per branch. They share E12, E23 and E56–E89. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

495 point of self-prediction. Restriction resonance, needed for the interpretation  
496 in Eq. (7), is a separate condition. The result binds the fixed point to the  
497 specified local states; compatibility and accuracy at that state are premises,  
498 while uniqueness follows from the construction.

499 The joint witnesses combine a double well, a register, a changing uniform-  
500 grid mesh and a three-site cortex; the active branch includes the noisy two-bit  
501 actuator. They establish that the hypotheses can hold together in distinct  
502 toy components. Mesh convergence is supplied independently of the thermo-  
503 dynamic bounds, and no common biological mechanism is established. The  
504 active witness rejects a zero heat budget and an incompatible energy limit.  
505 Counterexamples also expose the role of overlap agreement. The develop-  
506 ment declares no additional Lean axioms beyond the foundational axioms  
507 used by its libraries; physical premises are explicit hypotheses and model  
508 fields.

## 509 7 Discussion

510 The strongest results concern what a coherence measurement leaves unre-  
511 solved. Different recovery mechanisms share stationary distributions; equal  
512 phases permit incompatible local descriptions; and the tested plasticity ob-  
513 jective sustains coherence while moving away from its diagnostic structure.  
514 These findings make the coupling hypothesis more demanding to test and  
515 prevent a successful synchronization model from doing unexplained repre-

516 sentational work.

517     The proposed account identifies a compatible, internally represented state  
518 with a conscious episode. Its main explanatory gap is the mechanism that  
519 would maintain those relations: compatibility and accuracy at the glued state  
520 enter the construction as premises. Their mathematical consequences leave  
521 open why cortical activity should acquire or preserve them.

522     The most informative next step for that connection is a test of whether  
523 coordination helps maintain independently measured content agreement. Af-  
524 ter validating separate decoders and their overlap restrictions, a controlled  
525 change in coupling or coordination could test whether agreement deterio-  
526 rates in the predicted way, with controls for changes in sensory information  
527 and decoder quality. Such a result would bear directly on the proposed role  
528 of coherence. In parallel, the awakening hypothesis needs its own calibrated  
529 geometry–coupling relation and resolved phase dynamics to distinguish a cou-  
530 pling increase from diffusion change or common drive. A favorable recovery  
531 trace alone would leave the representational connection open.

532     Testing the self clause requires perturbing a candidate encoding and com-  
533 paring the resulting changes in represented scene, body and viewpoint with  
534 the readout’s predictions. Establishing that functional role would make dis-  
535 sociations from unified experience informative. The exact fixed-state ideal-  
536 ization must also be assessed against the fidelity and time dependence of the  
537 biological process.

538     Integrated-information, global-workspace and predictive-processing ac-

539 counts organize unity, access and modelling differently (Tononi et al., 2016;  
540 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal  
541 contributes a candidate interaction mechanism and explicit relations to test  
542 between local content and internal self-encoding. Its explanatory value will  
543 turn on whether those relations can be identified independently and shown  
544 to matter for experience. For the cup example, that means explaining how  
545 the system maintains one represented situation and how the perceiver’s place  
546 within it contributes to the episode.

## 547 **Data and code availability**

548 The Lean development, simulation scripts, tests and saved numerical sum-  
549 maries are maintained in the manuscript source repository. Publication sim-  
550 ulation values are regenerated from saved summaries without rerunning pro-  
551 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset  
552 ds005620 (Bajwa et al., 2025); preprocessing and selection are described  
553 in Supplemental Material. The source repository contains the reproducible  
554 analysis and does not redistribute the raw recordings.

## 555 **Funding and competing interests**

556 This research received no external funding. The author declares no compet-  
557 ing interests.

## 558 Use of AI tools

559 AI assistance to this work was substantial and is not separable by task.  
560 Several coding assistants and chat interfaces were used in an interleaved way  
561 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude  
562 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code  
563 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude  
564 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them  
565 they contributed to the Lean development, the simulation and analysis code,  
566 and the drafting and revision of both documents. Because the tools were  
567 used in combination and in alternation, no section, proof, figure or result is  
568 attributable to any one of them, and none is claimed to be. The author is  
569 responsible for all content, including everything a tool produced. Machine  
570 checking establishes the stated Lean results under their hypotheses; it does  
571 not validate biological interpretations or replace scientific review of generated  
572 code and prose.

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